

BTC Conversa - Your Personalized Banking Assistant

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Abstract

BTC Conversa: Generative AI Banking Chatbot is designed to simplify and automate mundane banking through the use of conversational AI. In this Phase One for the project, most of the concentration will be placed on the Information Bot, which greeting responses associated with account-related inquiries, balance inquiries, account statements, and branch information. The system is developed on Flask for API management and SQLite to simulate banking database management. It processes multiple queries via RASA for seamlessness, making the experience personalized. Also, session management by this bot maintains consistency across devices and browsers. This report describes the work completed during the first phase and underlines the chatbot's potential to bring about a sea change in customer interactions within the banking sector.

Chapter 1

Introduction

The banking sector has steadily moved towards digital platforms, enabling customers to manage their accounts, transfer funds, and pay bills online. However, many banking apps still require complex navigation and lack the user-friendliness needed for day-to-day banking tasks. In today's ever developing world, customers demand more personalized and faster solutions to simplify their interactions with banks.

This project, *BTC Conversa*, is designed to tackle these needs by introducing a conversational AI-based banking assistant. Using generative AI technology, this chatbot will automate routine banking tasks, enabling users to interact with their banks seamlessly through text-based conversations. This will allow users to complete tasks such as checking balances, managing accounts, and viewing transaction history without the need to navigate through a myriad of screens.

1.1 Problem Statement

Traditional banking systems and apps are functional but they fall short of delivering user-friendly experiences. Customers face delays in accessing important account information and frequently find the interface complicated to use. This leads to poor customer satisfaction, especially when users need personalized financial advice or quick access to their account details.

BTC Conversa aims to solve these issues by providing a single conversational AI-based solution. With features like real-time transaction history, automated responses to common queries, and personalized financial insights, BTC Conversa seeks to streamline banking for the modern day customer.

1.2 Motivation

This project is motivated by the high demand for better and easier to use banking services. In the age of digital banking adoption — requiring more personalized, real-time interactions from AI-powered solutions is the new normality.

The inspiration behind BTC conversa comes out of the requirement to give a more natural, human like experience to users. By incorporating conversational AI, we want to build a product not just to make banking tasks simpler but which also provides more personalized insights and support for an improved customer experience.

Additionally, this project is being carried out in collaboration with BluTech Consulting, providing valuable industry insights and guidance, ensuring that the solution aligns with real-world needs.

1.3 Scope

The scope of BTC Conversa defines a conversational AI-powered banking assistant to ease and automate routine banking tasks through natural language. This paper focuses, first of all, on major functionalities for the balance inquiry, account statement retrieval, money transfer, and branch information request. One can ask about several things in a single message, such as "What is my balance, ntn, and e-mail?" or ask for specific details about each transaction.

The system will be integrated seamlessly with banking for total security and personal experiences both on the web and mobile; BTC Conversa will also include session storage across device and browser changes to make users' data safe and consistent throughout their banking activities. It shall also be able to provide personalized financial insights to users based on their transaction history for better spending management and improved financial literacy.

BTC Conversa will allow pure financial and non-financial action handling, such as bill payments, fund transfers across accounts, actions related to adding a new payee, and spending alert management through text or voice-based conversations. Besides that, advisory services will also be provided regarding personalized tips for budgeting, saving on taxes, and spending patterns. It will also keep the records of all branch-related information updated within the system for facilitating the process of finding branches based on certain criteria related to region or availability on weekends.

In the future versions, more modules will be integrated to enable this chatbot with advanced analytics and advisory functions. BTC Conversa serves both personal banking customers and financial institutions by providing them with scalable solutions that are

adaptable to on-site banking needs.

1.4 Modules

BTC Conversa is divided into multiple modules, each handling specific banking functions. These modules include Information Management, Action Handling, Analysis, and Advisory Services, all integrated into a unified chatbot platform.

1.4.1 Module 1: Information Management

This module allows users to manage and access personal and account-related information.

1. Update personal information such as contact details, address, and security settings.
2. View and retrieve account details such as balance, account type, and transaction history.
3. Display information related to bank branches, including branch timings and services.

1.4.2 Module 2: Action Handling

This module manages both financial and non-financial actions for the user.

1. Perform financial actions such as bill payments, fund transfers, and adding payees.
2. Handle non-financial actions like card management, setting spending alerts, and filing complaints.

1.4.3 Module 3: Analysis Module

The analysis module helps users understand their spending and financial behavior.

1. Expenditure analysis based on transaction history to review spending patterns.
2. Payment analysis for insights into regular payments and financial habits.

1.4.4 Module 4: Advisory Module

This module offers personalized financial advice based on the user's transactions.

1. Personalized tips on managing finances, saving, and tax optimization.
2. Advisory on improving financial habits and managing personal budgets.

1.5 User Classes and Characteristics

BTC Conversa is designed for a variety of user classes, each with specific characteristics and needs. Below is a breakdown of the anticipated user types.

User Class	Description
Bank Customers	Bank customers will manage their accounts with BTC Conversa and query balance, initiate transfers, pay bills, and receive financial tips, all based on their relationships. The system is envisioned to be web and mobile-friendly for all sorts of variable devices; one session might see a customer executing several of these actions, including personalized queries that the chatbot would answer in real time.
Bank Staff	Bank staff will be given access to the system for the management of customer interactions and debugging of issues preventing the chatbot from operating effectively. They will have administrative access so that they can view and update backend information, such as details about any particular branch and also the system logs. They will also analyze the user data to improve the customer services and find trends.
IT Administrators	IT Administrators will be in charge of the infrastructure of the chatbot, ensuring security protocols, >95% uptime of the system, and resolution of any technical problems. They are also to deploy updates and manage API integrations with existing systems at the bank
Financial Advisors	With the analytics and advisory built into the chatbot, financial advisors can be privy to a customer's financial habits. Detailed financial data will then be at their disposal, when they can offer personalized recommendations to users through BTC Conversa.

Table 1.1: User Classes and Characteristics

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the BTC Conversa project.

2.1 Use-case/Event Response Table/Storyboarding

BTC Conversa is an interactive end-user application, so we will represent its requirements using a **Use Case Diagram** and a detailed **Use Case description** for each module. These use cases will highlight the interaction between users and the system, particularly in the areas of account information queries, fund transfers, and branch information retrieval.

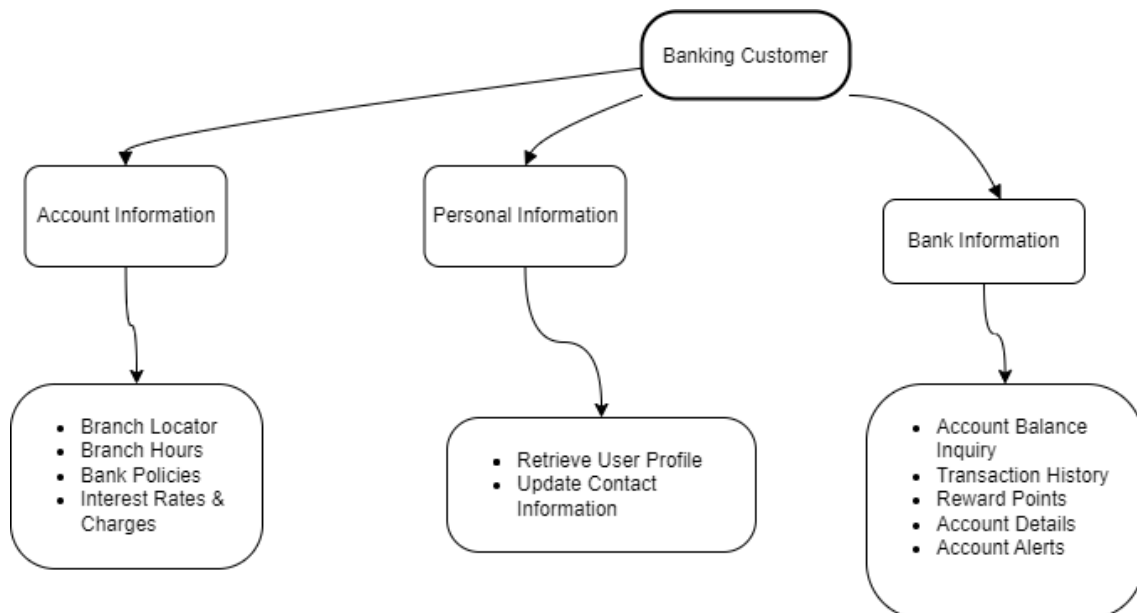


Figure 2.1: Use Case Diagram for BTC Conversa

2.1.1 Detailed Use Cases

Use Case 1: Retrieve Account Balance

Description: The user asks for their account balance, and the chatbot responds with the requested information.

Actors: User, System

Precondition: User is authenticated and has an active session.

Flow of Events:

- The user initiates the query by asking, "What is my account balance?"
- The system verifies the user's authentication.
- The system retrieves the balance from the database.
- The system returns the balance to the user in a conversational format.

Postcondition: The user receives the balance and the session remains active.

Use Case 2: View Transaction History

Description: The user requests their transaction history, and the chatbot provides a summary of transactions.

Actors: User, System

Precondition: User is authenticated and has requested a valid date range.

Flow of Events:

- The user asks, "Show me my transaction history for the last 30 days."
- The system verifies the user's authentication.
- The system retrieves the transaction history from the database.
- The system presents the transaction details to the user.

Postcondition: The user is presented with the requested transaction history.

2.2 Functional Requirements

This section describes the functional requirements of the system, organized by feature. Each module has specific functionalities that contribute to the overall user experience.

2.2.1 Module 1: Information Management

Following are the requirements for the Information Management module:

1. The system shall allow users to query account details, such as balance and account number.
2. The system shall allow users to view their transaction history for a specified period.
3. The system shall provide details about bank branches, including address, timings, and weekend availability.

2.2.2 Module 2: Action Handling

Following are the requirements for the Action Handling module:

1. The system shall allow users to transfer funds between accounts.
2. The system shall allow users to add new payees and manage payees.
3. The system shall allow users to pay bills through the chatbot interface.

2.3 Non-Functional Requirements

This section specifies non-functional requirements that will ensure the quality, performance, and security of the system.

2.3.1 Reliability

The system should have an availability rate of 99.9%, with a mean time between failures (MTBF) of 10,000 hours. Error detection should be automated, and the system should provide real-time error logs for debugging and recovery.

2.3.2 Usability

The system should be easy to use for both tech-savvy and non-tech-savvy users. The chatbot shall respond to user queries within 2-3 seconds and support natural language queries in both English and Urdu.

USE-1: The chatbot shall allow users to query multiple account details in a single interaction, such as "What is my balance and my recent transactions?"

2.3.3 Performance

Performance is critical to ensuring user satisfaction. The system should handle up to 1,000 concurrent users with no more than a 5% drop in response time.

PER-1: 95% of queries should be processed within 3 seconds of submission, assuming an Internet connection of at least 10 Mbps.

2.3.4 Security

The system must protect user data and banking information. Sensitive information such as account balances and transaction details should be encrypted in transit and at rest.

SEC-1: The system shall implement two-factor authentication (2FA) to ensure the security of user accounts. SEC-2: The system should log all access attempts and flag suspicious behavior for administrative review.

2.4 Domain Model

The domain model for BTC Conversa defines the key entities and their relationships in the system. The core entities include User, Account, Transaction, Branch, and Chatbot. These entities will be interconnected to support the system's functionality.

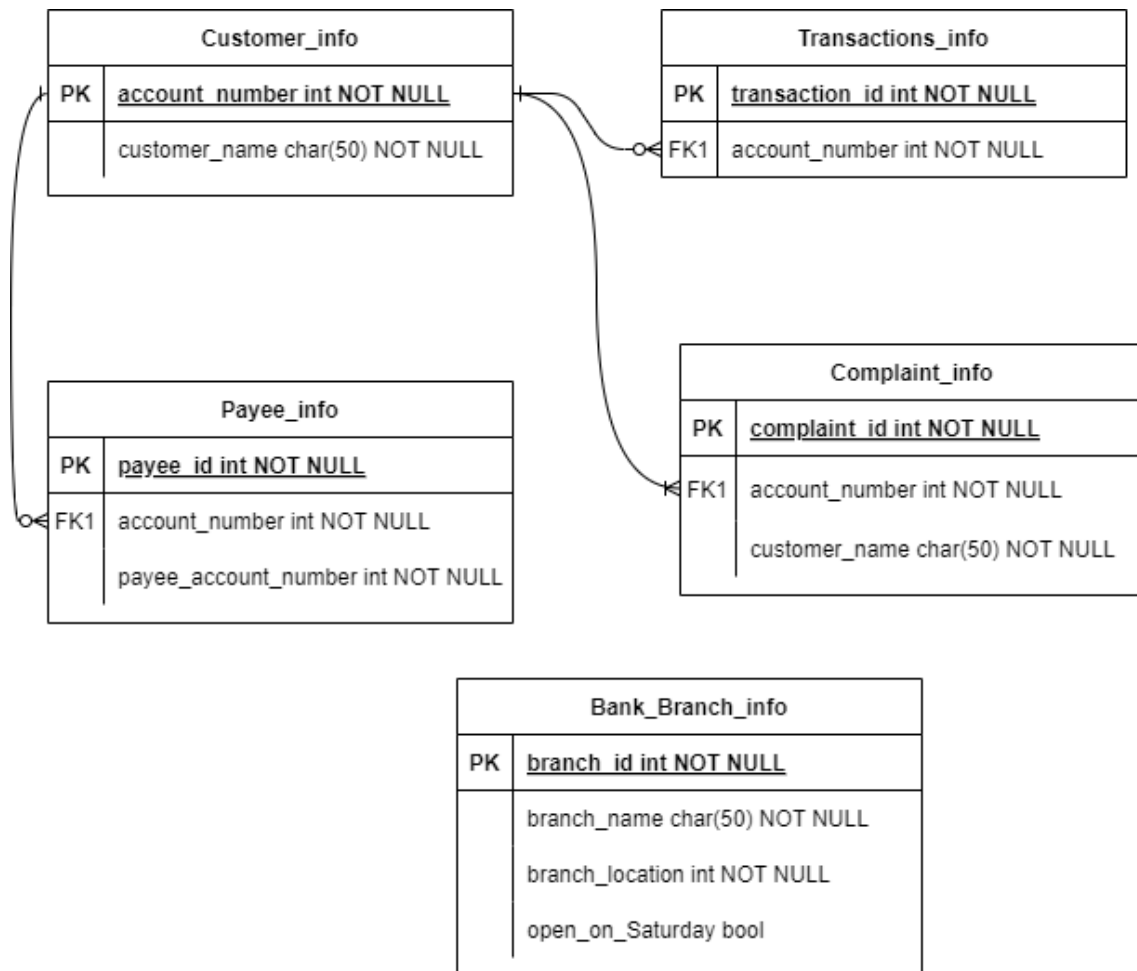


Figure 2.2: Domain Model for BTC Conversa

Chapter 3

System Overview

3.1 General Description

BTC Conversa is designed to be a conversational AI banking assistant and is envisioned to guide the users in performing routine banking tasks. This encapsulates most of the key functionalities, including account balance inquiries, transaction history, fund transfers, and requests for branch information. Users can interact with the system using a chatbot interface that processes real-time natural language input using RASA.

The architecture is client-server, where the interface of the chatbot interacts with the back-end services and APIs to fetch and process information related to banking. The system sustains security and usability for both web and mobile platforms by managing the sessions to ensure continuity between devices.

3.2 Architectural Design

The architecture of BTC Conversa follows a modular client-server design. The primary modules include:

- 1. Frontend:** Provides the chatbot interface for user interactions. Users can query information such as account balance, transaction history, and transfer funds through this interface.
- 2. API Layer (Flask):** Serves as the intermediary between the frontend and the backend systems. It handles API requests, authenticates users, and routes the queries to the appropriate service.
- 3. Backend Database (SQLite):** Stores essential banking data such as user accounts, transaction records, and branch information. The database is accessed via API calls.
- 4. Chatbot (RASA):** Handles natural language processing and intent recognition. It iden-

tifies the user's query and triggers the appropriate action or API call.

5. Session Management: Manages user sessions to ensure secure and personalized experiences. Sessions store authentication tokens and track user activity across devices.

6. LLaMA Model: For advanced user queries, such as financial analysis or spending habits, the LLaMA model is invoked to process data and generate insights.

Box and Line Diagram: The following diagram provides a high-level representation of the system's architecture, showing how the major modules interact with each other:

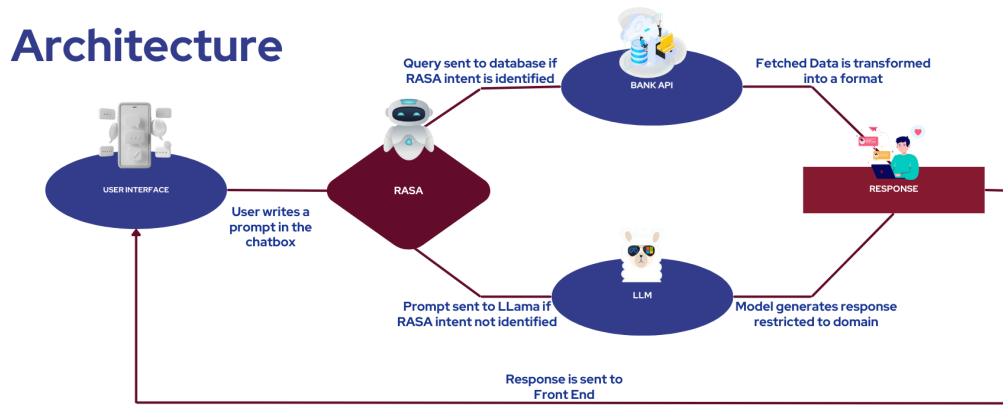


Figure 3.1: Box and Line Diagram of BTC Conversa Architecture

These subsystems are decompositions for scalability, modularity, and separation of concerns. Users' interactions are handled by the frontend, while complex data processing and storage happen in the backend; the API layer bridges the communication between the layers to make sure that the chatbot fetches data in real time to meet user requests.

3.3 Design Models

BTC Conversa utilizes the object-oriented development approach. The following design models are applicable:

The activity diagram represents the user's interaction with the chatbot, from login to query processing. It details the steps the system follows when handling a request, such as querying account balance or transaction history.

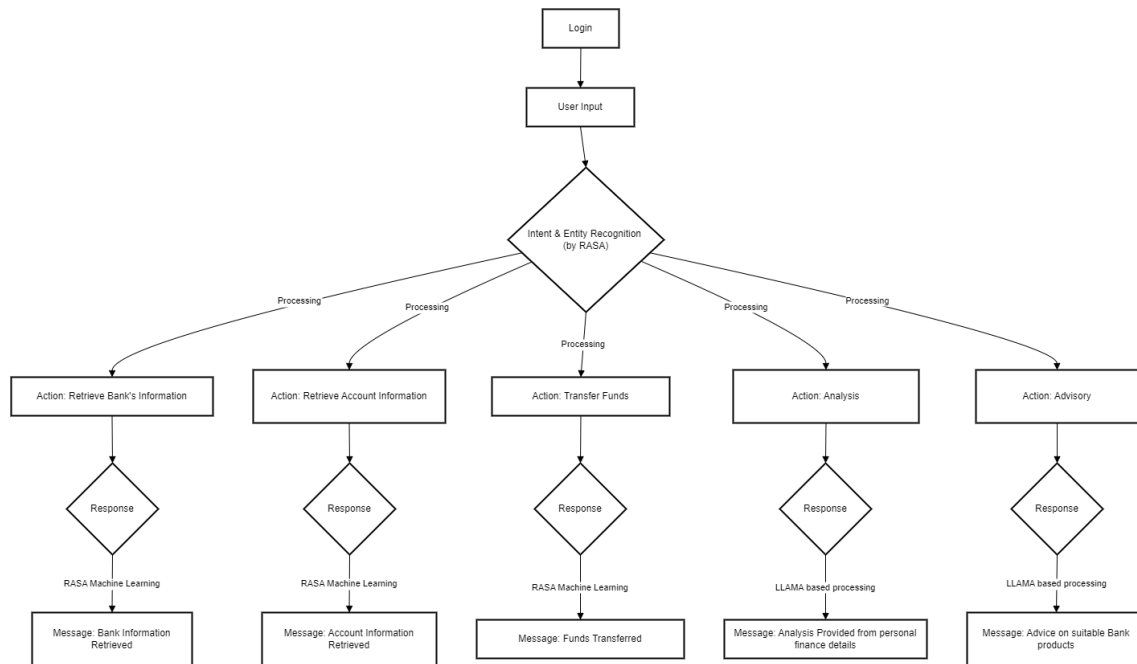


Figure 3.2: Activity Diagram for BTC Conversa

A class diagram was provided earlier in the report, showing the structure of key entities such as 'User', 'Account', 'Transaction', 'Chatbot', etc. This is a more generic view of our project and its structure.

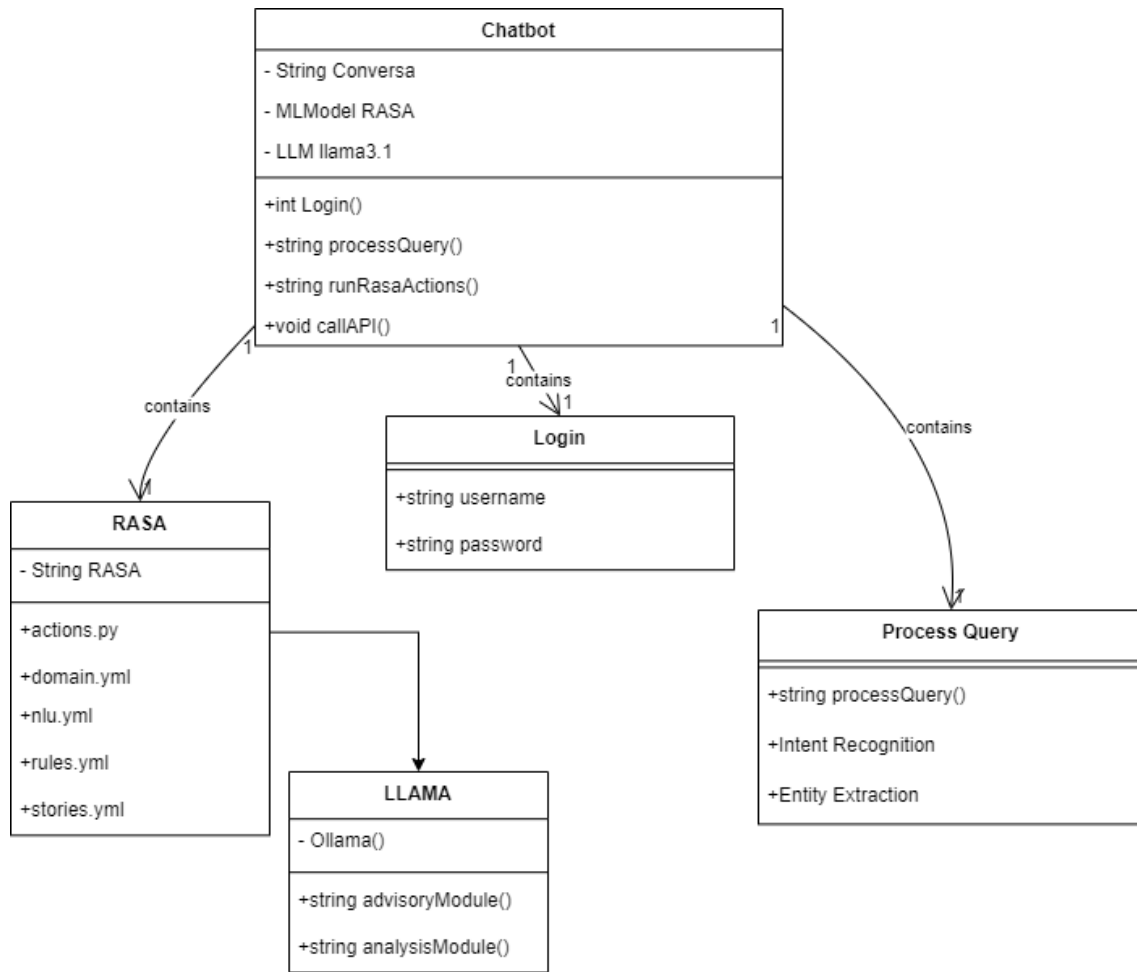


Figure 3.3: Simple Class Diagram for BTC Conversa

The sequence diagram shows the flow of information between objects and systems during an account balance inquiry. It traces the steps from user input, to API calls, to retrieving data from the backend, and finally delivering the response to the user.

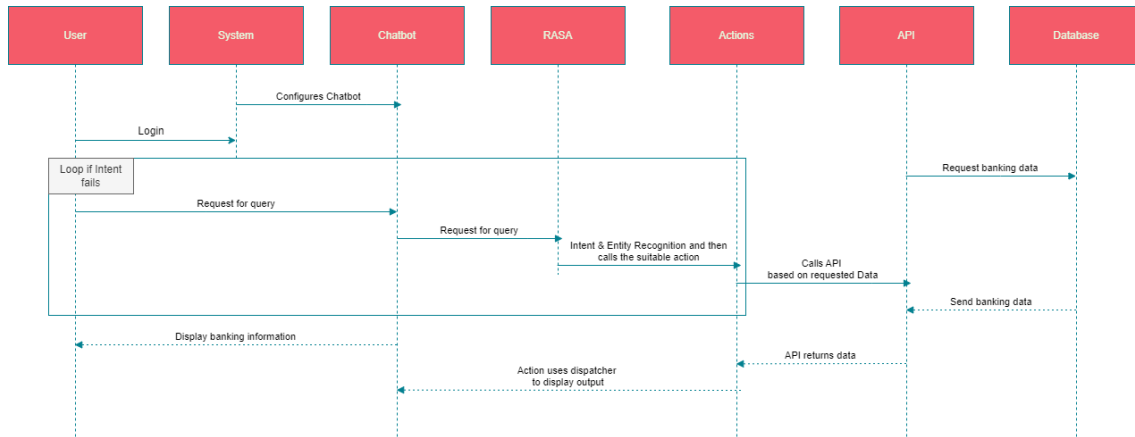


Figure 3.4: Sequence Diagram for BTC Conversa Query Processing

3.4 Data Design

The data design for BTC Conversa is focused on how user data, account information, and transaction history are stored and processed. The system uses a relational database (SQLite) to manage the banking data.

- **Users Table:** Contains user profile information including name, email, and authentication credentials.
- **Accounts Table:** Stores account details such as account number, balance, and the type of account (e.g., savings, current).
- **Transactions Table:** Logs all financial transactions made by the user, including the transaction date, amount, and associated account.
- **Branches Table:** Contains information about bank branches, such as the address, contact details, and availability on weekends.

Data Storage and Processing: - All user-related data is stored securely using encrypted storage. - The backend processes incoming queries by retrieving relevant information from the database and sends the processed data back to the frontend. - The system ensures secure communication through session tokens, which protect user data during interactions.

Chapter 4

Conclusions and Future Work

4.1 Conclusions

BTC Conversa is a conversational AI Banking Assistant engineered to be more interactive with the user's banking services. The initial part of the project involves setting up the fundamental architecture and introducing basic features such as balance inquiry, transaction history, and elementary fund transfers.

During this stage, the chatbot is at a partial implementation using RASA for natural language understanding and SQLite to simulate banking databases. It can process simple queries and respond in real time. Advanced functionality remains under development.

The project is in development with BluTech Consulting. The first version has created the bedrock, while further versions are needed to get close to the envisioned intelligent assistant for full assistance in banking. Initial progress is promising, and the modular architecture of the system ensures that future features can be integrated seamlessly.

Future stages of the project will be directed at improving overall functionality, usability, and security features of BTC Conversa. As this project progresses with development, it shall eventually become a fully capable and secure banking assistant that should appeal to the increasing demands of this modern banking industry.

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